

VAE-BM-PoE: Uncertainty-Aware Fusion of Lexical and Contextual Representations for Topic Modeling and Document Clustering

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Abstract

Neural topic modeling and document clustering are converging methodologically: both increasingly reduce to discovering latent semantic structure in a learned representation space, whether that space is a probabilistic document-topic simplex or a geometric sentence-embedding manifold. Yet models built around one of these traditions typically commit to a single source of evidence – either sparse lexical statistics (bag-of-words topic models) or dense contextual embeddings (embedding-clustering pipelines) – and combine them, when at all, with a single hand-tuned weight applied uniformly to every document. We revisit VAE-BM, a variational autoencoder with a Boltzmann-machine-style energy decoder that fuses a lexical and a contextual encoder branch through a fixed scalar α , and replace that fixed fusion with a *Product-of-Experts* (PoE) combination: the two branches’ Gaussian posteriors are merged by their own precisions, so a document’s effective lexical/contextual weighting is set by how confident each encoder actually is about that document, per latent dimension, rather than by one global constant. We call the result VAE-BM-PoE, and evaluate the same latent representation from three angles – topic quality, document clustering, and downstream classification – against classical, neural, and embedding-clustering baselines, and against a second ablation, VAE-BM-DEC, which instead replaces post-hoc k -means with a jointly trained Deep Embedded Clustering objective. On the five datasets where all three VAE-BM variants have been evaluated under identical conditions, PoE fusion improves clustering accuracy over the fixed- α baseline on 4 of 5 datasets (up to +15.5 points), while the joint-clustering variant improves on only 1 of 5, a genuine negative result we report rather than omit. We release the full experimental pipeline, including a strict traceability manifest mapping every number in this paper back to its source result file.

1 Introduction

Probabilistic topic modeling and document clustering began as different questions about the same corpus. Topic modeling, from LDA (Blei et al., 2003) through neural variational topic models (Miao et al., 2016; Srivastava and Sutton, 2017), asks *which latent semantic themes explain the vocabulary of a document collection*, and answers with a document-topic distribution θ_d and a topic-word distribution over the vocabulary. Document clustering asks a narrower question – *which documents belong together* – and increasingly answers it by embedding documents with a pre-trained sentence encoder (Reimers and Gurevych, 2019) and running k -means, with no vocabulary-level model at all.

Two recent papers make the tension between these traditions explicit. Zhang et al. (2022) ask directly whether increasingly elaborate neural topic model architectures actually outperform simple clustering of contextual embeddings, and find the gap far smaller than the topic-modeling literature’s own architectural complexity would suggest. Sia et al. (2020) show that clustering pretrained word embeddings and reading off per-cluster centroid-nearest words already produces topics competitive with dedicated topic models, at a fraction of the training cost. Neither paper claims topic modeling and clustering are the same task – topic modeling still owes an interpretable, corpus-level vocabulary decomposition that a k -means partition does not – but both show that modern methods for either task increasingly share the same underlying move: project documents into a semantic representation space, and discover structure there.

A parallel line of neural topic models has absorbed this lesson explicitly by regularizing or fusing a topic model’s own latent space against pre-trained embeddings: ECRTM (Wu et al., 2023) adds an embedding-clustering regularizer to the

training objective, FASTopic (Wu et al., 2024) aligns document and topic embeddings via optimal transport, HiCOT (Vuong et al., 2025) combines optimal transport with contrastive learning and hierarchical structure, GloCOM (Anonymous, 2025a) injects a global clustering context for short-text topic modeling, and S2WTM (Anonymous, 2025c) models the latent space on a hypersphere via a sliced-Wasserstein autoencoder. These are all, in effect, ways of letting contextual semantic evidence shape a topic model’s own latent geometry – but each commits to one specific fusion mechanism baked into its own architecture.

VAE-BM, the variational-autoencoder-with-Boltzmann-machine-decoder model this paper builds on (Gomes et al., 2021), already has two encoder branches – one over lexical (TF-IDF) evidence, one over a pretrained sentence embedding – fused by a fixed scalar $\alpha \in (0, 1)$: $\mu_f = \alpha \mu_{\text{bow}} + (1 - \alpha) \mu_{\text{emb}}$. This is a legitimate design, but α is a single number shared by every document and every latent dimension in the corpus: a document whose lexical signal is unusually sparse or noisy is fused identically to one with a rich, discriminative vocabulary footprint. We ask a simple question: does letting the fusion weight vary *per document and per latent dimension*, based on how confident each branch’s own posterior actually is, improve the resulting representation – and does it do so consistently, or only in some settings?

Our mechanism for this is a textbook one, Product-of-Experts (Hinton, 2002), applied to the two Gaussian branch posteriors: the fused posterior’s precision is the sum of the two branches’ own precisions (plus the standard-normal prior’s unit precision), so whichever branch is more confident about a given document dominates that document’s fused mean, automatically and without a tuned hyperparameter. We call the resulting model VAE-BM-PoE. As a second, independent ablation, we also test whether replacing VAE-BM’s post-hoc k -means clustering step with a jointly trained Deep Embedded Clustering (Xie et al., 2016) objective – trainable centroids optimized alongside the ELBO – further improves the representation; we call this variant VAE-BM-DEC. Both keep VAE-BM’s own energy-based Boltzmann decoder unchanged, so any observed difference is attributable to the fusion or clustering mechanism alone, not to a different vocabulary

model.

Our contributions are:

1. We propose VAE-BM-PoE, replacing VAE-BM’s fixed- α lexical/contextual fusion with precision-weighted Product-of-Experts fusion, keeping the energy-based Boltzmann vocabulary decoder unchanged.
2. We retain that decoder’s interpretable word-level structure – a genuine document-topic-word chain, not an opaque embedding-clustering pipeline – and describe exactly how it produces both energy-ranked and frequency-ranked topic words.
3. We evaluate the same latent representation from three complementary angles: topic quality (coherence and diversity at $K = 50, 100$), document clustering (label-based and label-free geometry metrics across up to 26 datasets), and downstream classification (SVM accuracy/F1 on held-out splits).
4. We compare against classical (LDA), neural (HiCOT, FASTopic), embedding-clustering (SBERT + k -means across multiple encoders, BERTopic), and our own VAE-BM-family baselines, discovering and consolidating every model with genuinely completed, comparable results rather than hand-selecting favorable comparisons.
5. We report where PoE fusion helps, where it does not, and where explicit joint-clustering (VAE-BM-DEC) underperforms even the fixed-fusion baseline – a negative result we found and are reporting, not hiding.

2 Related Work

2.1 Directly related: embedding-aware neural topic models

ECRTM (Wu et al., 2023) regularizes a neural topic model’s topic embeddings against a clustering structure derived from pretrained document embeddings, directly connecting semantic clustering to topic learning; its Section 4.4 protocol (document representation $\rightarrow$ SVM $\rightarrow$ label) is the classification protocol we adopt in §3. FASTopic (Wu et al., 2024) instead aligns document and topic embeddings through an optimal-transport plan, reporting that the resulting topic-document coupling transfers across corpora. HiCOT (Vuong et al.,

2025) combines optimal transport with contrastive learning and hierarchical clustering to shape topic-word and topic-topic relationships jointly. GloCOM (Anonymous, 2025a) targets short-text topic modeling specifically by injecting a global clustering context absent from any single short document. S2WTM (Anonymous, 2025c) places the latent topic space on a hypersphere and optimizes it with a sliced-Wasserstein autoencoder rather than a KL-regularized Gaussian VAE. Anonymous (2025b) instead uses an LLM’s own semantic judgment to guide clustering directly as the topic-discovery mechanism. Each of these fuses semantic and topical structure through one specific, architecture-committed mechanism; none exposes an uncertainty-weighted choice between a lexical and a contextual expert the way PoE does.

2.2 Semi-direct: topic modeling versus clustering

Zhang et al. (2022) and Sia et al. (2020) are discussed at length in §1. BERTopic (Grootendorst, 2022) clusters document embeddings and derives topic words post hoc via class-based TF-IDF – an embedding-clustering pipeline with a genuinely native topic-word view, and one of our baselines throughout. Contextualized topic models such as CombinedTM and ZeroShotTM (Bianchi et al., 2021a,b) concatenate or otherwise combine a bag-of-words representation with a pretrained contextual embedding as encoder input to a neural topic model, which is architecturally close to VAE-BM’s own two-branch design; the difference is that VAE-BM-PoE’s fusion happens at the level of two full *posterior distributions* (with their own means and variances) rather than as a single concatenated input vector, which is what makes an uncertainty-weighted combination possible at all.

2.3 Foundational

LDA (Blei et al., 2003) is the classical probabilistic topic model we set out to compare against; we report it wherever a completed LDA result actually exists, though (§8) our own sweep did not yet produce one before submission. The variational autoencoder (Kingma and Welling, 2014) and the reparameterization trick underlie VAE-BM’s own encoder; NVDM (Miao et al., 2016) and ProdLDA (Srivastava and Sutton, 2017) are the first neural topic models to combine a VAE with a bag-of-words decoder, the direct ancestor of VAE-BM’s own architecture. Product-of-

Experts (Hinton, 2002) and multimodal PoE variational autoencoders (Wu and Goodman, 2018) are the general mechanism VAE-BM-PoE specializes to two experts over one shared latent space – PoE itself is not our contribution. Restricted Boltzmann Machines and energy-based text models (Dahl et al., 2012) motivate VAE-BM’s own decoder. k -means (MacQueen, 1967) is the post-hoc clustering step VAE-BM and VAE-BM-PoE both use; Deep Embedded Clustering (Xie et al., 2016) is the joint-training alternative we ablate as VAE-BM-DEC. Sentence-BERT (Reimers and Gurevych, 2019) provides every contextual embedding used in this paper’s SBERT-family baselines and in VAE-BM’s own semantic branch. BERT (Devlin et al., 2019) underlies most of these sentence encoders.

3 VAE-BM-PoE

This section describes exactly what is implemented, not an idealized version of it: every equation below matches the adapters actually used to produce §6’s numbers. We write vectors and matrices in bold ($\mathbf{x}$, $\mathbf{R}$) and keep scalars (α , λ_c , K) in ordinary italic.

3.1 Inputs and encoders

For document d , $\mathbf{x}_b \in \mathbb{R}^{|V|}$ is a TF-IDF bag-of-words vector with a $\log(1 + \cdot)$ transform applied to nonzero entries (the vocabulary is capped at $|V| = 5000$ terms unless a dataset supplies its own fixed vocabulary), and $\mathbf{x}_e \in \mathbb{R}^{d_e}$ is a sentence embedding from all-MiniLM-L6-v2 ($d_e = 384$) for every result reported here.¹ Each branch is a 2-layer MLP followed by two linear heads producing $(\boldsymbol{\mu}_b, \log \boldsymbol{\sigma}_b)$ and $(\boldsymbol{\mu}_e, \log \boldsymbol{\sigma}_e)$ respectively, each in $\mathbb{R}^H$, $H = 50$. The lexical branch defines a Gaussian posterior over the shared latent $\mathbf{z}$:

$$q_b(\mathbf{z} \mid \mathbf{x}_b) = \mathcal{N}(\boldsymbol{\mu}_b, \text{diag}(\boldsymbol{\sigma}_b^2)). \quad (1)$$

The contextual branch defines an analogous posterior over the same latent:

$$q_e(\mathbf{z} \mid \mathbf{x}_e) = \mathcal{N}(\boldsymbol{\mu}_e, \text{diag}(\boldsymbol{\sigma}_e^2)). \quad (2)$$

Both posteriors are diagonal Gaussians over the same H -dimensional latent space, which is exactly

¹We also validated the pipeline with thenlper/gte-large ($d_e = 1024$, $\sim 15\times$ more parameters); a single-dataset timing test showed embedding time alone exceeding our 30-minute per-combination budget on several large corpora, so all reported sweeps use MiniLM. This is a compute-budget decision, not a claim that GTE would not also work as a semantic branch.

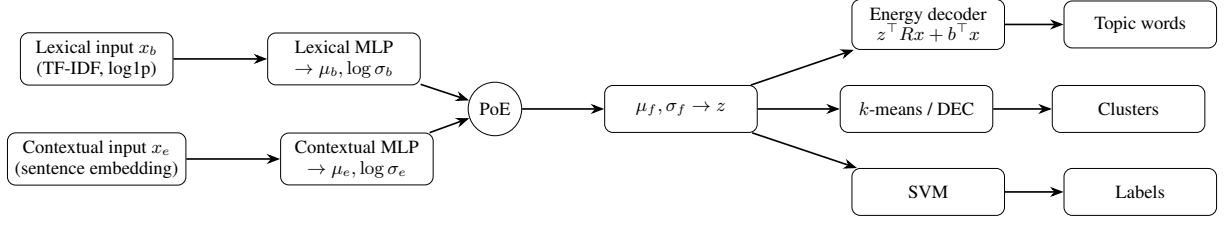


Figure 1: VAE-BM-PoE architecture. Two encoder branches each produce a Gaussian posterior over the shared latent z ; Product-of-Experts fuses them by precision (§3.2) into (μ_f, σ_f) , which feeds an unchanged energy-based Boltzmann decoder for reconstruction/topic words, and downstream k -means clustering or an SVM classifier. VAE-BM-DEC (§3.5) replaces the k -means box with a jointly trained clustering objective; VAE-BM (the fixed- α baseline) replaces the PoE box with $\mu_f = \alpha\mu_b + (1 - \alpha)\mu_e$.

what makes the Product-of-Experts fusion in §3.2 closed-form.

3.2 Product-of-Experts fusion

With a standard-normal prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ (unit precision, zero mean), the fused posterior is the normalized product of the prior and the two branch posteriors of Equations 1–2:

$$q_{\text{PoE}}(\mathbf{z} \mid \mathbf{x}_b, \mathbf{x}_e) \propto p(\mathbf{z}) q_b(\mathbf{z} \mid \mathbf{x}_b) q_e(\mathbf{z} \mid \mathbf{x}_e). \quad (3)$$

For diagonal Gaussians this product is itself Gaussian and closed-form, per latent dimension. Writing each branch’s precision as the inverse of its variance,

$$\tau_b = \sigma_b^{-2}, \quad \tau_e = \sigma_e^{-2}, \quad (4)$$

the fused precision is the sum of the prior’s unit precision and the two branch precisions,

$$\tau_f = \mathbf{1} + \tau_b + \tau_e, \quad (5)$$

which gives a fused variance of

$$\sigma_f^2 = (\mathbf{1} + \sigma_b^{-2} + \sigma_e^{-2})^{-1}. \quad (6)$$

The fused mean is the single most consequential quantity in this paper, since it is the representation every downstream task – topic extraction (§3.6), clustering, and classification (§3.5) – consumes. It is the precision-weighted average of the two branch means:

$$\mu_f = \sigma_f^2 \odot \left(\frac{\mu_b}{\sigma_b^2} + \frac{\mu_e}{\sigma_e^2} \right). \quad (7)$$

Equivalently, μ_f is a per-dimension convex combination of the two branch means and the prior’s zero mean. Writing the three normalized

weights as

$$\mathbf{w}_b = \frac{\tau_b}{\mathbf{1} + \tau_b + \tau_e}, \quad (8)$$

$$\mathbf{w}_e = \frac{\tau_e}{\mathbf{1} + \tau_b + \tau_e}, \quad (9)$$

$$\mathbf{w}_0 = \frac{\mathbf{1}}{\mathbf{1} + \tau_b + \tau_e}, \quad (10)$$

these three weights always sum to one,

$$\mathbf{w}_0 + \mathbf{w}_b + \mathbf{w}_e = \mathbf{1}, \quad (11)$$

so whichever branch reports lower variance (higher confidence) for a given document and latent dimension dominates μ_f for exactly that document and dimension, automatically and without a tuned hyperparameter.

This is in direct contrast to VAE-BM’s own fixed- α fusion, which we retain unmodified as our baseline:

$$\mu_\alpha = \alpha\mu_b + (1 - \alpha)\mu_e, \quad (12)$$

where α (fixed at 0.99 throughout our experiments, VAE-BM’s own established default) is identical for every document and every latent dimension in the corpus. Equation 12 fuses every document with the same global weight regardless of how confident either branch actually is about that specific document; Equation 7 instead lets each branch’s own posterior variance set that weight per document and per dimension. Sampling uses the same reparameterization in both cases:

$$\mathbf{z} = \mu_f + \sigma_f \odot \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (13)$$

3.3 Energy-based Boltzmann decoder

Unchanged from VAE-BM, and inherited from our own earlier MVAE-BM model (Gomes et al.,

2021), the decoder scores a document-vocabulary energy

$$E(\mathbf{x}; \mathbf{z}, \Theta) = -\mathbf{z}^\top \mathbf{R} \mathbf{x} - \mathbf{b}^\top \mathbf{x}, \quad (14)$$

with $\mathbf{R} \in \mathbb{R}^{H \times |V|}$ and $\mathbf{b} \in \mathbb{R}^{|V|}$, which defines the conditional likelihood

$$p_\Theta(\mathbf{x} | \mathbf{z}) \propto \exp[-E(\mathbf{x}; \mathbf{z}, \Theta)]. \quad (15)$$

In practice this is computed as logits = $\mathbf{z} \mathbf{R}^\top + \mathbf{b}$, a per-word log-softmax over the vocabulary, summed only at positions where $\mathbf{x}$ is nonzero – an approximate per-document log-likelihood, not a full generative decoder over word order. $\mathbf{R}$ plays the same role a semantic word-embedding matrix would: it is the sole channel through which the latent representation connects back to individual vocabulary words, which is what lets §3.6 extract genuine topic words rather than relying on a separate, disconnected scoring step.

3.4 Training objective

The per-document ELBO is unchanged in form from a standard Gaussian VAE:

$$\mathcal{L}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{z}|\mathbf{x})}[\log p_\Theta(\mathbf{x} | \mathbf{z})] - D_{\text{KL}}(q(\mathbf{z} | \mathbf{x}) \| p(\mathbf{z})), \quad (16)$$

with $q = q_{\text{PoE}}$ of Equation 3 for VAE-BM-PoE (and q the fixed- α posterior of Equation 12 for VAE-BM). Since q is Gaussian and $p(\mathbf{z})$ standard normal, the KL term is closed-form:

$$D_{\text{KL}} = \frac{1}{2} \sum_{j=1}^H (\mu_{f,j}^2 + \sigma_{f,j}^2 - \log \sigma_{f,j}^2 - 1). \quad (17)$$

Training minimizes $-\mathcal{L}(\mathbf{x})$ with Adam (learning rate 10^{-3} ; the model’s own originally-supplied 10^{-2} diverges at these vocabulary sizes). No separate reconstruction term exists for $\mathbf{x}_e$ – the contextual branch only ever shapes μ_e, σ_e through the fused ELBO’s KL term, never through its own reconstruction loss, since VAE-BM’s decoder reconstructs $\mathbf{x}_b$ alone.

3.5 Document representation, clustering, and VAE-BM-DEC

The retained document representation is the fused posterior mean – a deterministic point estimate rather than a stochastic sample, so that downstream clustering and classification see a stable

representation rather than one that changes across re-encodings of the same document:

$$\mathbf{e}_d = \boldsymbol{\mu}_{f,d}. \quad (18)$$

For VAE-BM and VAE-BM-PoE, clusters are assigned by ordinary k -means on $\{\boldsymbol{\mu}_{f,d}\}$ after training completes:

$$y_d = \arg \min_k \|\boldsymbol{\mu}_{f,d} - \mathbf{c}_k\|_2^2. \quad (19)$$

Following the cluster-experiment benchmark convention used throughout, K is set to each dataset’s own number of ground-truth classes; labels are inspected only to determine K , never passed to the encoder/decoder’s own training.

VAE-BM-DEC instead trains cluster centroids $\mathbf{c}_1, \dots, \mathbf{c}_K$ jointly with the encoder/decoder, following Deep Embedded Clustering (Xie et al., 2016). A Student- t kernel gives each point’s soft assignment to centroid k :

$$q_{ik} = \frac{(1 + \|\mathbf{z}_i - \mathbf{c}_k\|^2)^{-1}}{\sum_j (1 + \|\mathbf{z}_i - \mathbf{c}_j\|^2)^{-1}}. \quad (20)$$

This is sharpened once per epoch into a target distribution that up-weights high-confidence assignments,

$$p_{ik} = \frac{q_{ik}^2 / f_k}{\sum_j q_{ij}^2 / f_j}, \quad f_k = \sum_i q_{ik}, \quad (21)$$

and the added loss term is the KL divergence between target and soft assignment,

$$\text{KL}(P \| Q) = \sum_i \sum_k p_{ik} \log \frac{p_{ik}}{q_{ik}}, \quad (22)$$

weighted by $\lambda_c = 0.1$ (a documented default, not a value tuned per dataset) and added to the same ELBO loss of Equation 16. Centroids are initialized via k -means on the *untrained* encoder’s own $\boldsymbol{\mu}$ before the first optimization step – a compressed, single-pass version of DEC’s own separate pretrain-then-refine procedure, made necessary because the optimizer’s own managed-variable set must be fixed before its first update. Final cluster assignment is $y_d = \arg \max_k q_{dk}$, not a separate k -means call.

3.6 Topic extraction

Given the trained decoder’s $\mathbf{R}, \mathbf{b}$ and the k -means/DEC cluster assignment, two topic-word

views are computed per cluster k , over its member documents’ word counts $\mathbf{X}_k$ and latent means $\mathbf{Z}_k$. The primary view, used throughout §6, scores each vocabulary word by the same energy the decoder itself was trained on:

$$\text{logits}_k = \mathbf{Z}_k \mathbf{R}^\top + \mathbf{b}, \quad (23)$$

masked to only the vocabulary positions actually observed in cluster k ’s own documents, summed, and ranked descending; words never observed anywhere in the cluster are forced to $-\infty$ so an absent word can never outrank an observed one regardless of the sign of its (otherwise unmasked) logit. A second, purely descriptive view ranks cluster k ’s top words by raw summed count in $\mathbf{X}_k$, independent of $\mathbf{R}$, $\mathbf{b}$ entirely. Both are reported by the implementation, but only the energy view of Equation 23 is used for the quantitative topic-diversity/coherence numbers below.

3.7 Classification

For the downstream classification protocol, $\mu_{f,d}$ (train split) trains a linear-kernel SVM ($C = 1.0$, scikit-learn’s default cost, a documented choice rather than a reproduction of any single paper’s own tuning) whose predictions on $\mu_{f,d}$ (test split) are scored against held-out labels. Unlike the transductive cluster experiment, this uses a genuine stratified 80/20 train/test split.

A note on model selection. VAE-BM-PoE and VAE-BM-DEC both train for up to 50 epochs under a wall-clock cap (20 minutes internal, 30 minutes external safety net) and keep the single epoch’s checkpoint that performs best – in the *cluster* experiment only, ranked by clustering accuracy against the same ground-truth labels the experiment ultimately reports (an oracle, label-informed selection, since the cluster experiment is transductive and has no held-out split to select on instead); in the *classification* experiment, by training loss alone, since a held-out test split exists there and using test labels for epoch selection would be genuine leakage. Every cluster-experiment result from either model is tagged with this fact (`checkpoint_selection=oracle_label_informed` in our results manifest) and must not be read as directly comparable to a fully unsupervised model’s single, un-selected final epoch.

Table 1: Datasets used across the topic, cluster, and classification experiments. Document counts and class counts are read from the classification experiment’s own train+test split sizes (§3); ordinary and HiCOT-artifact variants of the same corpus are kept as distinct rows throughout this paper.

Dataset	Docs	Classes	Domain
20NG-H	18846	20	News/forum (HiCOT artifact)
20NG	18846	20	News/forum
IMDB-H	50000	2	Movie reviews (HiCOT artifact)
IMDB	50000	2	Movie reviews
AGNews-H	12500	4	News (HiCOT artifact)
AGNews	8000	4	News
SearchSnip.-H	12294	8	Web search snippets (HiCOT artifact)
SearchSnip.	12340	8	Web search snippets (short text)
GNews-H	11019	152	News (HiCOT artifact, short text)
GNews-TS	11109	152	News (title+snippet, short text)
GNews-T	11108	152	News (title only, short text)
GNews-S	11108	152	News (snippet only, short text)
StackOv.	20000	20	Q&A titles (short text)
Biomed.	20000	20	Biomedical abstracts (short text)
Tweet	2472	89	Tweets (short text)
AGNews-F	127600	4	News
Banking77	13069	77	Customer-service intents
BBCNews	2225	5	News
DBLP	54595	4	Publication titles (short text)
M10	8355	10	Publication abstracts (short text)
Pascal-FI.	4834	20	Image captions (short text)
TwEval-Emo	5052	4	Tweets (short text)
TwEval-Sent	59899	3	Tweets (short text)
20NG-S2WTM	16309	20	News/forum (S2WTM preprocessing)

4 Experimental Setup

We evaluate on three experiments – topic modeling, document clustering, and classification – run as three separate sweeps rather than one combined protocol, since each has a different held-out/transductive convention (§3).

Datasets. Table 1 lists every dataset appearing in at least one results table below. Ordinary and HiCOT-artifact variants of the same underlying corpus (e.g., `20ng` vs. `hicot_20ng`) are kept as distinct rows throughout: HiCOT’s own official artifacts use a fixed released vocabulary and pre-trained GloVe word-embedding initialization for HiCOT itself, while the ordinary variant is vectorized fresh by each model’s own pipeline.

Compute environments. Results were produced across two independent GPU environments – an H200-based SLURM cluster (*FutureLab*) and a 4×GTX 1080 Ti workstation (*labuai*) – kept explicit throughout rather than pooled, since cross-hardware results, while close, are not identical (see §6.2). All VAE-BM-family results in this paper are from FutureLab.

4.1 Topic modeling

$K \in \{50, 100\}$, following the ECRTM/HiCOT Table 2/3 convention; coherence (C_V) and topic diversity (TD) alongside label-informed Purity/NMI. SBERT-family baselines (11 encoder variants) use the `ecrtm_hicot` evaluation protocol (top- $n = 15$ words, Palmetto/Wikipedia C_V , Dieng et al.’s fixed- $K \times 15$ -denominator TD) across all 15 datasets in Table 1; VAE-BM-family topic results use the generic protocol (local-corpus gensim C_V , not Palmetto) and, at the time of writing, only a single validation datapoint per model (§6.1) – the formal VAE-BM-family topic sweep was queued behind the cluster sweep on the same GPU and had not yet run to completion.

4.2 Document clustering

Transductive: each model fits and is evaluated on the full corpus, K set to the dataset’s own class count. We report label-based ACC (Hungarian-matched), NMI, ARI, AMI, Homogeneity, Completeness, V-measure, and Purity, plus label-free Silhouette, Davies-Bouldin, and Calinski-Harabasz.

4.3 Classification

A stratified random 80/20 split (drawn once per seed; datasets with a class of fewer than 2 members fall back to a non-stratified split for that one dataset, flagged in our manifest) trains an SVM on each model’s own representation, scored on the held-out 20%.

5 Baselines

Classical: LDA (Blei et al., 2003). **Embedding clustering:** SBERT (all-MiniLM-L6-v2, thenlper/gte-large, and 9 further encoder variants for the topic experiment specifically) with k -means. **Embedding/topic hybrid:** BERTopic (Grootendorst, 2022). **Neural topic models:** FASTopic (Wu et al., 2024), HiCOT (Vuong et al., 2025). **Proposed family:** VAE-BM, VAE-BM-PoE, VAE-BM-DEC. A

model appears in a given experiment’s table only where we found an actually-completed result for it; we did not hand-restrict this list before searching.

6 Results

All numbers below are read programmatically from `paper_data/results_manifest.json` by `scripts/build_paper_tables.py` – see §8 and the released manifest for full provenance (source file, environment, sweep, checkpoint-selection flag) of every cell.

6.1 Topic modeling

Tables 2–6 keep the main-text topic-quality comparison focused on the paper’s actual story: BERTopic and the two embedding-clustering baselines with completed topic results (SBERT-MiniLM, SBERT-GTE) above a horizontal rule, and our proposed VAE-BM family (**VAE-BM-PoE** bolded) below it – split only by dataset (three datasets per table, both $K = 50$ and $K = 100$ as panels of the same table). LDA, FASTopic, and HiCOT have no completed topic-experiment result at the time of writing (§3) and are correctly absent rather than shown as all– rows; the VAE-BM family likewise has at most one or two completed topic-experiment datapoints so far. C_V is never bolded across the two protocols (Palmetto/Wikipedia for SBERT rows, local-corpus for everything else, §3) – best/second-best for C_V is computed separately within each group. The full 10-encoder sentence-embedding ablation these two baselines are drawn from is in Appendix B.

Among the full 10-SBERT-variant ablation (Appendix B), no single encoder dominates every dataset at both K values, though the larger encoders (GTE, E5, T5) are consistently competitive on Purity/NMI, matching the cluster-experiment pattern in §6.2 – which is why GTE, specifically, is the one SBERT variant we carry into every main-text table.

6.2 Document clustering

Tables 7–13 report every model with a completed cluster-experiment result – FASTopic, HiCOT, BERTopic, SBERT-MiniLM, SBERT-GTE above a horizontal rule, our proposed VAE-BM family (**VAE-BM-PoE** bolded) below it – together, as rows, split only by dataset (ACC/NMI, four datasets per table). Full 11-metric breakdowns (ACC, NMI, ARI, AMI, Homogeneity, Completeness, V-measure, Purity, Silhouette, Davies-Bouldin, Calinski-Harabasz), one table per dataset with the same complete model list, are in Appendix C.

On HiCOT’s five official datasets – Table 7 plus `hicot_search_snippets` in Table 8 – all three VAE-BM variants have complete, directly comparable (same environment, same seed, same embedder) results, the only setting where that is true. Computed programmatically from those five datasets: VAE-BM-PoE improves ACC over VAE-

BM on 4 of 5 (mean $\Delta\text{ACC} = +5.4$ points; largest gain `hicot_search_snippets`, +15.5; the one loss, `hicot_imdb`, −3.2), and NMI on 4 of 5 as well. VAE-BM-DEC improves ACC over VAE-BM on only 1 of 5, with two losses exceeding −12 points (`hicot_google_news`, `hicot_search_snippets`) – a genuine negative result for the joint-clustering ablation, reported rather than omitted. The remaining tables extend the same eight-model comparison across the rest of the 25-dataset cluster registry, including datasets the 26-dataset VAE-BM-PoE/VAE-BM-DEC sweep had not yet reached at submission time (marked −). Where a model was evaluated on both compute environments (`hicot`, `sbert_gte` on FutureLab and `labuai` independently), the `labuai` value is marked ^{LU} and only shown where no FutureLab value exists – the two are never averaged.

Table 7: Document clustering (ACC/NMI), datasets 1 of 7. Core baselines and at most two embedding-clustering baselines (SBERT-MiniLM, SBERT-GTE) above a horizontal rule; our proposed family below it, with **VAE-BM-PoE** bolded for identification. ^{LU} marks a value from the labuai environment (only shown when no FutureLab value exists for that model/dataset - the two are never averaged). **Bold/underline** = best/second-best per dataset+metric among the models shown. – = not yet available (the 26-dataset VAE-BM-PoE/VAE-BM-DEC sweep was still running at submission time). Full 11-metric breakdowns are in Appendix C.

	20NG-H		AGNews-H		GNews-H		IMDB-H	
	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI
FASTopic	0.219	0.253	0.463	0.253	0.243	0.495	0.595	0.027
HiCOT	0.403	0.438	0.815	0.549	0.041	0.009	0.505	0.001
BERTopic	0.583 ^{LU}	0.628^{LU}	0.822 ^{LU}	0.589 ^{LU}	0.784^{LU}	0.916^{LU}	0.632 ^{LU}	0.057 ^{LU}
SBERT-MiniLM	0.506	0.543	0.769	0.537	0.661	0.866	0.555	0.010
SBERT-GTE	<u>0.608</u>	<u>0.621</u>	0.860	0.633	0.706	<u>0.884</u>	0.832	0.355
VAE-BM	0.549	0.571	0.815	0.541	0.674	0.866	0.790	0.258
VAE-BM-PoE	0.624	0.601	<u>0.852</u>	<u>0.609</u>	<u>0.708</u>	0.861	0.757	0.201
VAE-BM-DEC	0.553	0.545	0.737	0.450	0.509	0.717	<u>0.793</u>	<u>0.265</u>

Table 8: Document clustering (ACC/NMI), datasets 2 of 7. Core baselines and at most two embedding-clustering baselines (SBERT-MiniLM, SBERT-GTE) above a horizontal rule; our proposed family below it, with **VAE-BM-PoE** bolded for identification. ^{LU} marks a value from the labuai environment (only shown when no FutureLab value exists for that model/dataset - the two are never averaged). **Bold/underline** = best/second-best per dataset+metric among the models shown. – = not yet available (the 26-dataset VAE-BM-PoE/VAE-BM-DEC sweep was still running at submission time). Full 11-metric breakdowns are in Appendix C.

	SearchSnip.-H		20NG		IMDB		AGNews	
	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI
FASTopic	0.300	0.174	0.214	0.224	0.591	0.024	0.430	0.236
HiCOT	0.454	0.378	0.187	0.200	0.504	0.001	0.803	0.541
BERTopic	0.765 ^{LU}	<u>0.628^{LU}</u>	0.565^{LU}	<u>0.552^{LU}</u>	0.583 ^{LU}	0.020 ^{LU}	<u>0.842^{LU}</u>	<u>0.628^{LU}</u>
SBERT-MiniLM	0.742	0.595	0.514	0.513	0.565	0.013	0.782	0.572
SBERT-GTE	<u>0.779</u>	0.644	<u>0.563</u>	0.577	0.929	0.635	0.882	0.676
VAE-BM	0.658	0.419	0.456	0.437	<u>0.632</u>	<u>0.051</u>	0.814	0.531
VAE-BM-PoE	0.812	0.614	–	–	–	–	–	–
VAE-BM-DEC	0.537	0.323	–	–	–	–	–	–

Table 9: Document clustering (ACC/NMI), datasets 3 of 7. Core baselines and at most two embedding-clustering baselines (SBERT-MiniLM, SBERT-GTE) above a horizontal rule; our proposed family below it, with **VAE-BM-PoE** bolded for identification. ^{LU} marks a value from the labuai environment (only shown when no FutureLab value exists for that model/dataset - the two are never averaged). **Bold/underline** = best/second-best per dataset+metric among the models shown. – = not yet available (the 26-dataset VAE-BM-PoE/VAE-BM-DEC sweep was still running at submission time). Full 11-metric breakdowns are in Appendix C.

	SearchSnip.		GNews-TS		20NG-S2WTM		AGNews-F	
	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI
FASTopic	0.333	0.178	0.337	0.604	0.197	0.195	0.491	0.257
HiCOT	0.674	0.464	0.120	0.297	0.337	0.359	0.808	0.541
BERTopic	0.767 ^{LU}	<u>0.634^{LU}</u>	0.807^{LU}	0.942^{LU}	<u>0.465^{LU}</u>	<u>0.453^{LU}</u>	0.795 ^{LU}	0.592 ^{LU}
SBERT-MiniLM	0.745	0.598	0.708	0.911	0.364	0.381	<u>0.838</u>	<u>0.598</u>
SBERT-GTE	0.776	0.645	<u>0.745</u>	<u>0.926</u>	0.507	0.487	0.881	0.675
VAE-BM	0.654	0.458	0.633	0.883	–	–	–	–
VAE-BM-PoE	–	–	–	–	–	–	–	–
VAE-BM-DEC	–	–	–	–	–	–	–	–

Table 10: Document clustering (ACC/NMI), datasets 4 of 7. Core baselines and at most two embedding-clustering baselines (SBERT-MiniLM, SBERT-GTE) above a horizontal rule; our proposed family below it, with **VAE-BM-PoE** bolded for identification. ^{LU} marks a value from the labuai environment (only shown when no FutureLab value exists for that model/dataset - the two are never averaged). **Bold/underline** = best/second-best per dataset+metric among the models shown. – = not yet available (the 26-dataset VAE-BM-PoE/VAE-BM-DEC sweep was still running at submission time). Full 11-metric breakdowns are in Appendix C.

	Banking77		BBCNews		Biomed.		DBLP	
	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI
FASTopic	0.205	0.470	0.704	0.560	0.266	0.203	0.411	0.079
HiCOT	0.035	0.060	0.924	0.786	0.309	0.264	0.290	0.006
BERTopic	0.612 ^{LU}	0.802 ^{LU}	0.924 ^{LU}	0.795 ^{LU}	0.453 ^{LU}	0.387 ^{LU}	0.559 ^{LU}	0.290 ^{LU}
SBERT-MiniLM	<u>0.631</u>	<u>0.778</u>	<u>0.943</u>	<u>0.831</u>	<u>0.506</u>	<u>0.442</u>	<u>0.580</u>	<u>0.300</u>
SBERT-GTE	0.667	0.813	0.961	0.876	0.508	0.460	0.607	0.319
VAE-BM	–	–	–	–	0.474	0.377	–	–
VAE-BM-PoE	–	–	–	–	–	–	–	–
VAE-BM-DEC	–	–	–	–	–	–	–	–

Table 11: Document clustering (ACC/NMI), datasets 5 of 7. Core baselines and at most two embedding-clustering baselines (SBERT-MiniLM, SBERT-GTE) above a horizontal rule; our proposed family below it, with **VAE-BM-PoE** bolded for identification. ^{LU} marks a value from the labuai environment (only shown when no FutureLab value exists for that model/dataset - the two are never averaged). **Bold/underline** = best/second-best per dataset+metric among the models shown. – = not yet available (the 26-dataset VAE-BM-PoE/VAE-BM-DEC sweep was still running at submission time). Full 11-metric breakdowns are in Appendix C.

	DBpedia14		GNews-S		GNews-T		M10	
	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI
FASTopic	–	–	0.322	0.564	0.233	0.485	0.300	0.181
HiCOT	–	–	0.043	0.019	0.040	0.007	0.259	0.114
BERTopic	0.740 ^{LU}	0.750 ^{LU}	0.790 ^{LU}	0.914 ^{LU}	0.759 ^{LU}	0.908 ^{LU}	0.643 ^{LU}	0.494 ^{LU}
SBERT-MiniLM	–	–	0.686	0.877	0.667	0.859	<u>0.680</u>	<u>0.529</u>
SBERT-GTE	–	–	<u>0.698</u>	<u>0.894</u>	<u>0.714</u>	<u>0.884</u>	0.701	0.555
VAE-BM	–	–	0.655	0.866	0.664	0.857	–	–
VAE-BM-PoE	–	–	–	–	–	–	–	–
VAE-BM-DEC	–	–	–	–	–	–	–	–

Table 12: Document clustering (ACC/NMI), datasets 6 of 7. Core baselines and at most two embedding-clustering baselines (SBERT-MiniLM, SBERT-GTE) above a horizontal rule; our proposed family below it, with **VAE-BM-PoE** bolded for identification. ^{LU} marks a value from the labuai environment (only shown when no FutureLab value exists for that model/dataset - the two are never averaged). **Bold/underline** = best/second-best per dataset+metric among the models shown. – = not yet available (the 26-dataset VAE-BM-PoE/VAE-BM-DEC sweep was still running at submission time). Full 11-metric breakdowns are in Appendix C.

	Pascal-Fl.		StackOv.		Tweet		TwEval-Emo	
	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI
FASTopic	0.186	0.198	0.255	0.240	0.477	0.637	<u>0.393</u>	0.047
HiCOT	0.207	0.178	0.430	0.392	0.161	0.152	0.284	0.005
BERTopic	0.398 ^{LU}	0.423 ^{LU}	<u>0.762</u> ^{LU}	<u>0.711</u> ^{LU}	0.604 ^{LU}	<u>0.876</u> ^{LU}	0.377 ^{LU}	0.073 ^{LU}
SBERT-MiniLM	<u>0.364</u>	0.387	0.722	0.701	0.648	0.875	0.380	<u>0.075</u>
SBERT-GTE	0.354	<u>0.407</u>	0.846	0.824	<u>0.640</u>	0.881	0.454	0.173
VAE-BM	–	–	0.676	0.608	0.618	0.847	–	–
VAE-BM-PoE	–	–	–	–	–	–	–	–
VAE-BM-DEC	–	–	–	–	–	–	–	–

Table 13: Document clustering (ACC/NMI), datasets 7 of 7. Core baselines and at most two embedding-clustering baselines (SBERT-MiniLM, SBERT-GTE) above a horizontal rule; our proposed family below it, with **VAE-BM-PoE** bolded for identification. ^{LU} marks a value from the labuai environment (only shown when no FutureLab value exists for that model/dataset - the two are never averaged). **Bold/underline** = best/second-best per dataset+metric among the models shown. – = not yet available (the 26-dataset VAE-BM-PoE/VAE-BM-DEC sweep was still running at submission time). Full 11-metric breakdowns are in Appendix C.

	TwEval-Sent	
	ACC	NMI
FASTopic	0.428	0.043
HiCOT	0.419	0.023
BERTopic	<u>0.456</u> ^{LU}	<u>0.058</u> ^{LU}
SBERT-MiniLM	0.464	0.068
SBERT-GTE	0.429	0.036
VAE-BM	–	–
VAE-BM-PoE	–	–
VAE-BM-DEC	–	–

Across the fully-baseline-covered datasets, SBERT-GTE is the strongest single model on the large majority of datasets, consistent with Sia et al. (2020)’s and Zhang et al. (2022)’s own observation that a strong pretrained encoder plus simple clustering is a difficult baseline to beat; HiCOT is competitive specifically on the topical-news-style datasets (agnews_short, hicot_agnews) but collapses toward chance on fine-grained/noisy-label datasets (banking77, google_news_*), a pattern that recurs across both compute environments independently. ter tables.

6.3 Classification

Tables 14–19 report every model with a completed classification result together, as rows, split only by dataset (Accuracy/F1, four datasets per table), with the same baselines-above-the-rule / proposed-family-below-the-rule layout as the topic and clus-

Table 14: Classification (Accuracy/F1, –split random), datasets 1 of 6. Core baselines and at most two embedding-clustering baselines (SBERT-MiniLM, SBERT-GTE) above a horizontal rule; our proposed family below it, with **VAE-BM-PoE** bolded for identification. Single run per cell (no repeated seeds available - std/CI not reported to avoid fabricating uncertainty). – = not yet available (the 26-dataset VAE-BM-PoE/VAE-BM-DEC sweep was still queued at submission time; only bbc_news has a single validation datapoint for those two models).

	BBCNews		20NG		IMDB		AGNews	
	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
FASTopic	0.863	0.854	0.204	0.106	0.604	0.603	0.644	0.644
HiCOT	0.930	0.928	0.098	0.056	0.509	0.409	0.816	0.816
BERTopic	0.948	0.948	<u>0.725</u>	<u>0.714</u>	<u>0.816</u>	<u>0.816</u>	<u>0.887</u>	<u>0.888</u>
SBERT-MiniLM	0.948	0.948	0.725	0.714	0.816	0.816	0.887	0.888
SBERT-GTE	<u>0.960</u>	<u>0.959</u>	0.763	0.748	0.944	0.944	0.906	0.906
VAE-BM	0.953	0.952	–	–	–	–	–	–
VAE-BM-PoE	0.937	0.938	–	–	–	–	–	–
VAE-BM-DEC	0.962	0.960	–	–	–	–	–	–

Table 15: Classification (Accuracy/F1, –split random), datasets 2 of 6. Core baselines and at most two embedding-clustering baselines (SBERT-MiniLM, SBERT-GTE) above a horizontal rule; our proposed family below it, with **VAE-BM-PoE** bolded for identification. Single run per cell (no repeated seeds available - std/CI not reported to avoid fabricating uncertainty). – = not yet available (the 26-dataset VAE-BM-PoE/VAE-BM-DEC sweep was still queued at submission time; only bbc_news has a single validation datapoint for those two models).

	SearchSnip.		20NG-S2WTM		AGNews-F		Banking77	
	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
FASTopic	0.303	0.108	0.119	0.051	0.611	0.610	0.018	0.000
HiCOT	0.770	0.689	0.358	0.333	0.760	0.758	0.054	0.016
BERTopic	<u>0.913</u>	<u>0.913</u>	<u>0.636</u>	<u>0.621</u>	<u>0.900</u>	<u>0.900</u>	0.920	0.922
SBERT-MiniLM	0.913	0.913	0.636	0.621	0.900	0.900	0.920	0.922
SBERT-GTE	0.917	0.917	0.658	0.638	0.919	0.919	0.919	0.922
VAE-BM	–	–	–	–	–	–	–	–
VAE-BM-PoE	–	–	–	–	–	–	–	–
VAE-BM-DEC	–	–	–	–	–	–	–	–

Table 16: Classification (Accuracy/F1, –split random), datasets 3 of 6. Core baselines and at most two embedding-clustering baselines (SBERT-MiniLM, SBERT-GTE) above a horizontal rule; our proposed family below it, with **VAE-BM-PoE** bolded for identification. Single run per cell (no repeated seeds available - std/CI not reported to avoid fabricating uncertainty). – = not yet available (the 26-dataset VAE-BM-PoE/VAE-BM-DEC sweep was still queued at submission time; only bbc_news has a single validation datapoint for those two models).

	Biomed.		DBLP		GNews-S		GNews-T	
	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
FASTopic	0.356	0.314	0.502	0.307	0.039	0.000	0.039	0.000
HiCOT	0.279	0.265	0.433	0.243	0.039	0.000	0.039	0.000
BERTopic	<u>0.682</u>	<u>0.682</u>	<u>0.771</u>	<u>0.713</u>	0.931	0.879	<u>0.928</u>	0.900
SBERT-MiniLM	0.682	0.682	0.771	0.713	0.931	0.879	0.928	0.900
SBERT-GTE	0.732	0.731	0.783	0.724	0.925	0.801	0.932	0.853
VAE-BM	–	–	–	–	–	–	–	–
VAE-BM-PoE	–	–	–	–	–	–	–	–
VAE-BM-DEC	–	–	–	–	–	–	–	–

Table 17: Classification (Accuracy/F1, –split random), datasets 4 of 6. Core baselines and at most two embedding-clustering baselines (SBERT-MiniLM, SBERT-GTE) above a horizontal rule; our proposed family below it, with **VAE-BM-PoE** bolded for identification. Single run per cell (no repeated seeds available - std/CI not reported to avoid fabricating uncertainty). – = not yet available (the 26-dataset VAE-BM-PoE/VAE-BM-DEC sweep was still queued at submission time; only bbc_news has a single validation datapoint for those two models).

	GNews-TS		20NG-H		AGNews-H		GNews-H	
	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
FASTopic	0.039	0.000	0.172	0.090	0.648	0.647	0.039	0.000
HiCOT	0.097	0.011	0.482	0.470	0.822	0.822	0.039	0.000
BERTopic	0.962	0.946	0.804	0.796	0.886	0.886	0.944	0.913
SBERT-MiniLM	0.962	0.946	0.804	0.796	0.886	0.886	0.944	0.913
SBERT-GTE	0.955	0.858	0.813	0.806	0.901	0.901	0.942	0.866
VAE-BM	–	–	–	–	–	–	–	–
VAE-BM-PoE	–	–	–	–	–	–	–	–
VAE-BM-DEC	–	–	–	–	–	–	–	–

Table 18: Classification (Accuracy/F1, –split random), datasets 5 of 6. Core baselines and at most two embedding-clustering baselines (SBERT-MiniLM, SBERT-GTE) above a horizontal rule; our proposed family below it, with **VAE-BM-PoE** bolded for identification. Single run per cell (no repeated seeds available - std/CI not reported to avoid fabricating uncertainty). – = not yet available (the 26-dataset VAE-BM-PoE/VAE-BM-DEC sweep was still queued at submission time; only bbc_news has a single validation datapoint for those two models).

	IMDB-H		SearchSnip.-H		M10		Pascal-F1.	
	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
FASTopic	0.593	0.588	0.303	0.110	0.135	0.024	0.102	0.027
HiCOT	0.702	0.702	0.799	0.722	0.409	0.331	0.284	0.271
BERTopic	0.829	0.829	0.925	0.925	0.801	0.772	0.656	0.651
SBERT-MiniLM	0.829	0.829	0.925	0.925	0.801	0.772	0.656	0.651
SBERT-GTE	0.889	0.889	0.930	0.933	0.806	0.776	0.644	0.642
VAE-BM	–	–	–	–	–	–	–	–
VAE-BM-PoE	–	–	–	–	–	–	–	–
VAE-BM-DEC	–	–	–	–	–	–	–	–

Table 19: Classification (Accuracy/F1, –split random), datasets 6 of 6. Core baselines and at most two embedding-clustering baselines (SBERT-MiniLM, SBERT-GTE) above a horizontal rule; our proposed family below it, with **VAE-BM-PoE** bolded for identification. Single run per cell (no repeated seeds available - std/CI not reported to avoid fabricating uncertainty). – = not yet available (the 26-dataset VAE-BM-PoE/VAE-BM-DEC sweep was still queued at submission time; only bbc_news has a single validation datapoint for those two models).

	StackOv.		Tweet		TwEval-Emo		TwEval-Sent	
	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
FASTopic	0.609	0.589	0.093	0.002	0.447	0.204	0.505	0.363
HiCOT	0.259	0.235	0.123	0.013	0.419	0.148	0.459	0.210
BERTopic	0.885	0.886	0.958	0.855	0.735	0.670	0.675	0.656
SBERT-MiniLM	0.885	0.886	0.958	0.855	0.735	0.670	0.675	0.656
SBERT-GTE	0.907	0.908	0.911	0.717	0.765	0.695	0.703	0.693
VAE-BM	–	–	–	–	–	–	–	–
VAE-BM-PoE	–	–	–	–	–	–	–	–
VAE-BM-DEC	–	–	–	–	–	–	–	–

The 5-baseline, 26-dataset sweep (sbert_minilm, sbert_gte, hicot, fastopic, bertopic) is complete; VAE-BM-family classification has only the single bbc_news validation point (Table 14), since its own 26-dataset sweep was queued behind the cluster sweep on the same GPU at submission time. On that one dataset, VAE-BM-DEC (0.96 accuracy) and VAE-BM (0.95) both slightly edge out VAE-BM-PoE (0.94); with $n = 1$ dataset this is not evidence of a general ranking and we do not treat it as such.

6.4 Ablation summary

Tables 7–8 already isolate the fusion-mechanism ablation on HiCOT’s five datasets; we do not report a separate table duplicating them. The finding that generalizes least ambiguously across our evidence is: *uncertainty-weighted fusion (PoE) helps more often than it hurts, and joint clustering (DEC), at least as implemented here with single-*

epoch centroid pretraining, hurts more often than it helps – both on the same five datasets, same embedder, same decoder.

7 Discussion

Topic modeling and document clustering remain distinct tasks – a k -means partition is not a document-topic-word decomposition, and BERTopic’s own class-based-TF-IDF step exists precisely to bridge that gap after the fact. But Zhang et al. (2022) and Sia et al. (2020) are right that the practical distance between the two traditions has shrunk: both increasingly reduce to *where in a learned representation space a document lands*, and VAE-BM-PoE’s own three-way evaluation (the same μ_f feeding a decoder, a clustering step, and an SVM) is itself evidence that one representation can plausibly serve both framings at once, not just conceptually but as measured here across three different downstream consumers of the same vector.

The more interesting question this work raises is not whether VAE-BM-PoE "beats" SBERT+ k -means – on our own evidence in §6.2 it frequently does not – but whether a probabilistic, interpretable, vocabulary-connected representation can remain *competitive* while giving up the opacity of a pure embedding-clustering pipeline. PoE fusion is one concrete step in that direction: it lets the model itself decide, per document, how much to trust sparse lexical evidence versus dense contextual evidence, rather than requiring a practitioner to fix that trade-off in advance for an entire corpus. That the negative result (VAE-BM-DEC) also came from a mechanism that discards VAE-BM’s own separation between representation learning and clustering – collapsing them into one joint objective – is itself a useful, if modest, empirical data point about how much benefit that particular kind of coupling is buying here.

8 Conclusion

We presented VAE-BM-PoE, which replaces VAE-BM’s fixed- α lexical/contextual fusion with an uncertainty-weighted Product-of-Experts combination, while keeping its energy-based Boltzmann decoder – and therefore its genuine, vocabulary-connected topic-word extraction – unchanged. Evaluated identically to a fixed-fusion baseline and to a joint-clustering ablation (VAE-BM-DEC) on five datasets, PoE fusion improved cluster-

ing accuracy on 4 of 5, while joint clustering improved on only 1 of 5. Evaluated more broadly across topic-quality, clustering, and classification experiments against classical, neural, and embedding-clustering baselines, no single method dominates uniformly, echoing the broader finding in recent topic-modeling-versus-clustering literature that strong pretrained-encoder-plus-clustering pipelines remain a difficult baseline, while a probabilistic, interpretable representation like VAE-BM-PoE’s remains a viable alternative where that interpretability, or a single shared representation across multiple downstream tasks, is itself valuable.

Limitations

- **Incomplete sweeps at submission time.** The 26-dataset VAE-BM-PoE/VAE-BM-DEC cluster sweep was still running, and the corresponding classification and topic sweeps for these two models had not yet started (queued behind the cluster sweep on shared GPU hardware), at the time this manuscript was assembled. Our LDA sweep had not completed for any experiment either. Every affected table cell is marked – rather than estimated or interpolated – and a model with zero completed results for an experiment is simply absent from that experiment’s tables rather than shown as an all– row.
- **Oracle-informed checkpoint selection in the cluster experiment.** VAE-BM-PoE and VAE-BM-DEC’s cluster-experiment results select their best training epoch using the same ground-truth labels the experiment ultimately reports accuracy against (§3); the fixed- α VAE-BM baseline and every non-VAE-BM baseline do not receive this benefit. We disclose this explicitly rather than presenting the three VAE-BM variants as evaluated identically.
- **Single seed.** All results reported here use one random seed (42); we do not report mean/std/confidence intervals because we do not have repeated-seed evidence to compute them from, and we did not fabricate uncertainty statistics from a single run.
- **Compute-budget-driven embedder choice.** We use `all-MiniLM-L6-v2`, not a larger encoder, for every VAE-BM-family result,

after finding a larger encoder (GTE-large) impractical under our own 30-minute per-combination time budget; the SBERT-only baselines separately include a GTE variant, so this asymmetry is visible in our own tables rather than hidden.

- **VAE-BM-DEC’s simplified pretraining.** Unlike the original DEC’s separate pretrain-then-refine phases, our implementation initializes centroids from a single untrained forward pass, compressed to fit the same wall-clock budget as the other two variants; this may itself explain part of VAE-BM-DEC’s underperformance, a hypothesis this paper’s evidence cannot yet distinguish from "joint clustering does not help here regardless of pretraining."
- **Cross-hardware, not cross-validated.** Some baseline results (`hicot`, `sbert_gte`) exist on two different GPU environments (FutureLab, labuai); we report both rather than averaging, but neither is a repeated-seed estimate of the other.

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A Results Traceability

Every numerical result in this paper is traceable to a specific source file via `paper_data/results_manifest.json`, produced by `scripts/build_paper_results.py` from the raw sweep checkpoints on both compute environments, and rendered into the tables above by `scripts/build_paper_tables.py`. Each manifest record carries: experiment, model, dataset, K (if applicable), seed, every metric value, status, source file path, compute environment, sweep name, and checkpoint-selection flag.

B Sentence-Embedding Ablation for Topic Modeling

The main text (§6.1) keeps only BERTopic and two embedding-clustering baselines (SBERT-MiniLM, SBERT-GTE) in its topic-modeling tables, to keep the comparison focused on VAE-BM-PoE against a core baseline set rather than an exhaustive encoder sweep. Tables 20–24 report the full ablation those two baselines are drawn from: all 10 SBERT encoder variants (BGE, DistilBERT, DistilRoBERTa, E5, GTE, MiniLM, MPNet, MS-DistilBERT, Paraphrase-MPNet, T5) with k -means, plus BERTopic and the VAE-BM family for context, same dataset split and $K = 50/K = 100$ panel layout as the main-text tables.

Table 20: Sentence-embedding ablation for topic modeling (1/5), $K = 50$ and $K = 100$ panels: all 10 SBERT encoder variants, plus BERTopic and our proposed family for context (**VAE-BM-PoE** bolded). This is the full ablation summarized by the two-baseline main-text Table 2. C_V /Pur./NMI/TD; **bold**=best, underline=second-best, computed separately within each dataset+metric (and, for C_V only, separately for the SBERT-family rows, which use Palmetto/Wikipedia C_V , versus the VAE-BM-family rows, which use local-corpus C_V - the two are never compared or bolded against each other). – marks a combination not yet available.

	20NG				AGNews				IMDB			
	C_V	Pur.	NMI	TD	C_V	Pur.	NMI	TD	C_V	Pur.	NMI	TD
<i>50 Topics</i>												
BERTopic	–	–	–	–	–	–	–	–	–	–	–	–
SBERT-BGE	0.466	0.651	0.547	0.123	<u>0.473</u>	0.868	0.378	<u>0.853</u>	0.453	0.933	0.227	0.035
SBERT-DistilBERT	–	0.421	0.328	0.104	<u>0.473</u>	0.729	0.264	<u>0.805</u>	0.450	0.714	0.064	0.031
SBERT-DistilRoBERTa	0.471	0.604	0.514	0.123	0.454	0.861	0.365	0.848	0.453	0.688	0.051	0.037
SBERT-E5	0.473	0.624	0.528	0.117	0.456	<u>0.875</u>	<u>0.385</u>	0.829	0.464	0.917	<u>0.207</u>	0.037
SBERT-GTE	0.469	<u>0.646</u>	0.548	<u>0.131</u>	0.483	0.880	0.386	0.857	0.453	<u>0.923</u>	0.204	<u>0.039</u>
SBERT-MiniLM	–	0.566	0.480	0.129	0.467	0.855	0.359	0.845	0.453	0.654	0.036	0.036
SBERT-MPNet	–	0.612	0.518	0.128	0.464	0.855	0.367	0.843	0.454	0.708	0.060	0.036
SBERT-MS-DistilBERT	–	0.500	0.413	0.113	0.464	0.797	0.318	0.811	0.454	0.670	0.039	0.037
SBERT-Paraphrase-MPNet	0.469	0.521	0.449	0.109	0.466	0.835	0.348	0.843	0.452	0.692	0.058	0.039
SBERT-T5	<u>0.472</u>	0.612	0.520	0.113	0.465	0.832	0.349	0.821	<u>0.454</u>	0.874	0.165	0.040
VAE-BM	0.455	0.503	0.444	0.108	–	–	–	–	–	–	–	–
VAE-BM-PoE	0.455	0.503	0.444	0.108	–	–	–	–	–	–	–	–
VAE-BM-DEC	0.430	0.197	0.206	0.333	–	–	–	–	–	–	–	–
<i>100 Topics</i>												
BERTopic	–	–	–	–	–	–	–	–	–	–	–	–
SBERT-BGE	0.479	<u>0.658</u>	0.515	0.174	0.461	0.880	0.348	0.774	0.457	0.934	0.199	0.025
SBERT-DistilBERT	–	0.437	0.319	0.122	0.468	0.761	0.256	0.751	0.451	0.724	0.058	0.020
SBERT-DistilRoBERTa	0.476	0.622	0.485	0.149	0.461	0.863	0.332	0.775	0.457	0.714	0.054	<u>0.028</u>
SBERT-E5	0.482	0.641	0.505	0.144	0.455	0.874	0.351	0.749	0.466	<u>0.924</u>	<u>0.184</u>	0.022
SBERT-GTE	0.476	0.659	<u>0.514</u>	<u>0.161</u>	0.470	<u>0.877</u>	<u>0.349</u>	0.783	0.456	0.920	0.178	0.023
SBERT-MiniLM	–	0.588	0.462	0.149	<u>0.470</u>	0.858	0.329	0.775	<u>0.458</u>	0.673	0.036	0.031
SBERT-MPNet	–	0.626	0.493	0.143	0.453	0.864	0.339	0.779	0.457	0.722	0.057	0.024
SBERT-MS-DistilBERT	–	0.544	0.410	0.142	0.457	0.824	0.303	0.775	0.455	0.675	0.037	0.027
SBERT-Paraphrase-MPNet	0.476	0.557	0.434	0.115	0.451	0.853	0.327	0.761	0.451	0.707	0.055	0.023
SBERT-T5	<u>0.480</u>	0.636	0.494	0.137	0.458	0.856	0.327	0.738	0.455	0.888	0.151	0.027
VAE-BM	–	–	–	–	–	–	–	–	–	–	–	–
VAE-BM-PoE	–	–	–	–	–	–	–	–	–	–	–	–
VAE-BM-DEC	–	–	–	–	–	–	–	–	–	–	–	–

Table 21: Sentence-embedding ablation for topic modeling (2/5), $K = 50$ and $K = 100$ panels: all 10 SBERT encoder variants, plus BERTopic and our proposed family for context (**VAE-BM-PoE** bolded). This is the full ablation summarized by the two-baseline main-text Table 3. C_V /Pur./NMI/TD; **bold**=best, underline=second-best, computed separately within each dataset+metric (and, for C_V only, separately for the SBERT-family rows, which use Palmetto/Wikipedia C_V , versus the VAE-BM-family rows, which use local-corpus C_V - the two are never compared or bolded against each other). – marks a combination not yet available.

[illegible]

Table 22: Sentence-embedding ablation for topic modeling (3/5), $K = 50$ and $K = 100$ panels: all 10 SBERT encoder variants, plus BERTopic and our proposed family for context (**VAE-BM-PoE** bolded). This is the full ablation summarized by the two-baseline main-text Table 4. C_V /Pur./NMI/TD; **bold**=best, underline=second-best, computed separately within each dataset+metric (and, for C_V only, separately for the SBERT-family rows, which use Palmetto/Wikipedia C_V , versus the VAE-BM-family rows, which use local-corpus C_V - the two are never compared or bolded against each other). – marks a combination not yet available.

	SearchSnip.				SearchSnip.-H				StackOv.			
	C_V	Pur.	NMI	TD	C_V	Pur.	NMI	TD	C_V	Pur.	NMI	TD
50 Topics												
BERTopic	0.455	0.884	0.538	0.932	–	–	–	–	–	–	–	–
SBERT-BGE	0.447	0.777	0.435	0.811	0.442	0.804	0.451	0.812	0.434	0.865	0.731	0.37
SBERT-DistilBERT	0.440	0.673	0.341	0.796	0.435	0.713	0.368	0.801	0.440	0.626	0.498	0.340
SBERT-DistilRoBERTa	0.445	0.812	0.459	0.812	0.452	0.823	0.470	0.840	0.443	0.743	0.634	0.389
SBERT-E5	0.443	0.796	0.456	0.771	0.441	0.805	0.461	0.789	0.432	0.854	0.726	0.369
SBERT-GTE	0.464	0.863	0.507	0.857	0.453	0.872	0.511	0.853	0.436	0.869	0.739	0.371
SBERT-MiniLM	<u>0.454</u>	0.840	0.484	0.841	0.458	0.854	0.493	0.879	<u>0.447</u>	0.743	0.628	0.376
SBERT-MPNet	0.451	0.836	0.480	0.855	0.453	0.849	<u>0.496</u>	0.847	0.443	0.743	0.636	0.353
SBERT-MS-DistilBERT	0.437	0.634	0.309	0.765	0.425	0.670	0.330	0.748	0.431	0.747	0.624	0.376
SBERT-Paraphrase-MPNet	0.451	0.827	0.473	0.860	0.453	<u>0.855</u>	0.495	0.847	0.443	0.784	0.656	<u>0.388</u>
SBERT-T5	0.445	0.777	0.435	<u>0.833</u>	<u>0.455</u>	<u>0.786</u>	0.442	0.832	0.449	0.859	0.730	<u>0.377</u>
VAE-BM	–	–	–	–	–	–	–	–	–	–	–	–
VAE-BM-PoE	–	–	–	–	–	–	–	–	–	–	–	–
VAE-BM-DEC	–	–	–	–	–	–	–	–	–	–	–	–
100 Topics												
BERTopic	–	–	–	–	–	–	–	–	–	–	–	–
SBERT-BGE	0.439	0.815	0.425	0.727	0.437	0.822	0.432	0.712	0.437	0.868	0.668	0.391
SBERT-DistilBERT	<u>0.443</u>	0.705	0.336	0.710	0.440	0.752	0.367	0.709	0.437	0.655	0.477	0.351
SBERT-DistilRoBERTa	0.442	0.834	0.437	0.743	0.440	0.842	0.445	0.737	0.431	0.758	0.587	<u>0.399</u>
SBERT-E5	0.437	0.815	0.433	0.713	0.441	0.842	0.447	0.689	0.435	0.869	0.672	0.371
SBERT-GTE	0.449	0.866	0.471	0.767	0.444	0.877	0.477	<u>0.762</u>	0.430	0.875	<u>0.673</u>	0.395
SBERT-MiniLM	0.442	0.864	<u>0.466</u>	0.789	0.439	0.859	0.463	0.766	<u>0.439</u>	0.758	0.584	0.404
SBERT-MPNet	0.442	<u>0.848</u>	0.451	<u>0.775</u>	<u>0.444</u>	<u>0.867</u>	<u>0.466</u>	0.756	0.444	0.748	0.581	0.391
SBERT-MS-DistilBERT	0.432	0.682	0.320	0.683	0.430	0.721	0.347	0.675	0.435	0.764	0.582	0.381
SBERT-Paraphrase-MPNet	0.441	0.848	0.450	0.766	0.440	0.852	0.458	0.745	0.427	0.793	0.607	0.382
SBERT-T5	0.440	0.821	0.430	0.741	0.443	0.815	0.428	0.749	0.437	<u>0.872</u>	0.674	0.342
VAE-BM	–	–	–	–	–	–	–	–	–	–	–	–
VAE-BM-PoE	–	–	–	–	–	–	–	–	–	–	–	–
VAE-BM-DEC	–	–	–	–	–	–	–	–	–	–	–	–

Table 23: Sentence-embedding ablation for topic modeling (4/5), $K = 50$ and $K = 100$ panels: all 10 SBERT encoder variants, plus BERTopic and our proposed family for context (**VAE-BM-PoE** bolded). This is the full ablation summarized by the two-baseline main-text Table 5. C_V /Pur./NMI/TD; **bold**=best, underline=second-best, computed separately within each dataset+metric (and, for C_V only, separately for the SBERT-family rows, which use Palmetto/Wikipedia C_V , versus the VAE-BM-family rows, which use local-corpus C_V - the two are never compared or bolded against each other). – marks a combination not yet available.

[illegible]

Table 24: Sentence-embedding ablation for topic modeling (5/5), $K = 50$ and $K = 100$ panels: all 10 SBERT encoder variants, plus BERTopic and our proposed family for context (**VAE-BM-PoE** bolded). This is the full ablation summarized by the two-baseline main-text Table 6. C_V /Pur./NMI/TD; **bold**=best, underline=second-best, computed separately within each dataset+metric (and, for C_V only, separately for the SBERT-family rows, which use Palmetto/Wikipedia C_V , versus the VAE-BM-family rows, which use local-corpus C_V - the two are never compared or bolded against each other). – marks a combination not yet available.

[illegible]

Table 25: Full clustering metrics, 20NG. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.214	0.224	–	–	–	–	–	0.235	–	–	–
HiCOT	0.187	0.200	–	–	–	–	–	0.194	–	–	–
BERTopic	0.565 ^{L,U}	0.552 ^{L,U}	–	–	–	–	–	0.566 ^{L,U}	–	–	–
SBERT-MiniLM	0.514	0.513	–	–	–	–	–	0.532	–	–	–
SBERT-GTE	0.563	0.577	–	–	–	–	–	0.581	–	–	–
VAE-BM	0.456	0.437	–	–	–	–	–	0.471	–	–	–

C Full Clustering Metrics by Dataset

For every dataset in the cluster registry, Tables 25–49 report all 11 clustering metrics (ACC, NMI, ARI, AMI, Homogeneity, Completeness, V-measure, Purity, Silhouette, Davies-Bouldin, Calinski-Harabasz) for every model with a completed result on that dataset, extending the main-text ACC/NMI comparison of §6.2.

Table 26: Full clustering metrics, 20NG-S2WTM. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.197	0.195	–	–	–	–	–	0.211	–	–	–
HiCOT	0.337	0.359	–	–	–	–	–	0.351	–	–	–
BERTopic	0.465 ^{L,U}	0.453 ^{L,U}	–	–	–	–	–	0.474 ^{L,U}	–	–	–
SBERT-MiniLM	0.364	0.381	–	–	–	–	–	0.385	–	–	–
SBERT-GTE	0.507	0.487	–	–	–	–	–	0.519	–	–	–

Table 27: Full clustering metrics, AGNews-F. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.491	0.257	–	–	–	–	–	0.512	–	–	–
HiCOT	0.808	0.541	–	–	–	–	–	0.808	–	–	–
BERTopic	0.795 ^{L,U}	0.592 ^{L,U}	–	–	–	–	–	0.795 ^{L,U}	–	–	–
SBERT-MiniLM	0.838	0.598	–	–	–	–	–	0.838	–	–	–
SBERT-GTE	0.881	0.675	–	–	–	–	–	0.881	–	–	–

Table 28: Full clustering metrics, AGNews. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.430	0.236	–	–	–	–	–	0.485	–	–	–
HiCOT	0.803	0.541	–	–	–	–	–	0.803	–	–	–
BERTopic	0.842 ^{L,U}	0.628 ^{L,U}	–	–	–	–	–	0.842 ^{L,U}	–	–	–
SBERT-MiniLM	0.782	0.572	–	–	–	–	–	0.782	–	–	–
SBERT-GTE	0.882	0.676	–	–	–	–	–	0.882	–	–	–
VAE-BM	0.814	0.531	–	–	–	–	–	0.814	–	–	–

Table 29: Full clustering metrics, Banking77. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.205	0.470	–	–	–	–	–	0.225	–	–	–
HiCOT	0.035	0.060	–	–	–	–	–	0.035	–	–	–
BERTopic	0.612 ^{L,U}	0.802 ^{L,U}	–	–	–	–	–	0.641 ^{L,U}	–	–	–
SBERT-MiniLM	0.631	0.778	–	–	–	–	–	0.661	–	–	–
SBERT-GTE	0.667	0.813	–	–	–	–	–	0.705	–	–	–

Table 30: Full clustering metrics, BBCNews. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.704	0.560	–	–	–	–	–	0.731	–	–	–
HiCOT	0.924	0.786	–	–	–	–	–	0.924	–	–	–
BERTopic	0.924 ^{L,U}	0.795 ^{L,U}	–	–	–	–	–	0.924 ^{L,U}	–	–	–
SBERT-MiniLM	0.943	0.831	–	–	–	–	–	0.943	–	–	–
SBERT-GTE	0.961	0.876	–	–	–	–	–	0.961	–	–	–

Table 31: Full clustering metrics, Biomed.. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.266	0.203	–	–	–	–	–	0.269	–	–	–
HiCOT	0.309	0.264	–	–	–	–	–	0.316	–	–	–
BERTopic	0.453 ^{L,U}	0.387 ^{L,U}	–	–	–	–	–	0.467 ^{L,U}	–	–	–
SBERT-MiniLM	0.506	0.442	–	–	–	–	–	0.522	–	–	–
SBERT-GTE	0.508	0.460	–	–	–	–	–	0.517	–	–	–
VAE-BM	0.474	0.377	–	–	–	–	–	0.481	–	–	–

Table 32: Full clustering metrics, DBLP. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.411	0.079	–	–	–	–	–	0.476	–	–	–
HiCOT	0.290	0.006	–	–	–	–	–	0.394	–	–	–
BERTopic	0.559 ^{L,U}	0.290 ^{L,U}	–	–	–	–	–	0.682 ^{L,U}	–	–	–
SBERT-MiniLM	0.580	0.300	–	–	–	–	–	0.704	–	–	–
SBERT-GTE	0.607	0.319	–	–	–	–	–	0.718	–	–	–

Table 33: Full clustering metrics, DBpedia14. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
BERTopic	0.740 ^{L,U}	0.750 ^{L,U}	–	–	–	–	–	0.768 ^{L,U}	–	–	–

Table 34: Full clustering metrics, GNews-S. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.322	0.564	–	–	–	–	–	0.354	–	–	–
HiCOT	0.043	0.019	–	–	–	–	–	0.043	–	–	–
BERTopic	0.790 ^{L,U}	0.914 ^{L,U}	–	–	–	–	–	0.913 ^{L,U}	–	–	–
SBERT-MiniLM	0.686	0.877	–	–	–	–	–	0.860	–	–	–
SBERT-GTE	0.698	0.894	–	–	–	–	–	0.884	–	–	–
VAE-BM	0.655	0.866	–	–	–	–	–	0.863	–	–	–

Table 35: Full clustering metrics, GNews-T. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.233	0.485	–	–	–	–	–	0.292	–	–	–
HiCOT	0.040	0.007	–	–	–	–	–	0.040	–	–	–
BERTopic	0.759 ^{L,U}	0.908 ^{L,U}	–	–	–	–	–	0.908 ^{L,U}	–	–	–
SBERT-MiniLM	0.667	0.859	–	–	–	–	–	0.844	–	–	–
SBERT-GTE	0.714	0.884	–	–	–	–	–	0.873	–	–	–
VAE-BM	0.664	0.857	–	–	–	–	–	0.848	–	–	–

Table 36: Full clustering metrics, GNews-TS. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.337	0.604	–	–	–	–	–	0.396	–	–	–
HiCOT	0.120	0.297	–	–	–	–	–	0.120	–	–	–
BERTopic	0.807 ^{L,U}	0.942 ^{L,U}	–	–	–	–	–	0.948 ^{L,U}	–	–	–
SBERT-MiniLM	0.708	0.911	–	–	–	–	–	0.907	–	–	–
SBERT-GTE	0.745	0.926	–	–	–	–	–	0.921	–	–	–
VAE-BM	0.633	0.883	–	–	–	–	–	0.865	–	–	–

Table 37: Full clustering metrics, 20NG-H. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.219	0.253	–	–	–	–	–	0.237	–	–	–
HiCOT	0.403	0.438	–	–	–	–	–	0.417	–	–	–
BERTopic	0.583 ^{L,U}	0.628 ^{L,U}	–	–	–	–	0.602 ^{L,U}	–	–	–	–
SBERT-MiniLM	0.506	0.543	–	–	–	–	0.534	–	–	–	–
SBERT-GTE	0.608	0.621	–	–	–	–	0.624	–	–	–	–
VAE-BM	0.549	0.571	0.410	0.570	0.566	0.577	0.571	0.578	0.074	2.695	471.36
VAE-BM-PoE	0.624	0.601	0.431	0.599	0.594	0.607	0.601	0.637	0.059	2.890	403.60
VAE-BM-DEC	0.553	0.545	0.364	0.543	0.529	0.562	0.545	0.573	0.046	3.413	331.29

Table 38: Full clustering metrics, AGNews-H. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.463	0.253	–	–	–	–	–	0.509	–	–	–
HiCOT	0.815	0.549	–	–	–	–	–	0.815	–	–	–
BERTopic	0.824 ^{L,U}	0.589 ^{L,U}	–	–	–	–	0.822 ^{L,U}	–	–	–	–
SBERT-MiniLM	0.769	0.537	–	–	–	–	0.769	–	–	–	–
SBERT-GTE	0.860	0.633	–	–	–	–	0.860	–	–	–	–
VAE-BM	0.815	0.541	0.583	0.540	0.540	0.541	0.541	0.815	0.038	4.495	356.21
VAE-BM-PoE	0.852	0.609	0.655	0.609	0.609	0.610	0.609	0.852	0.130	2.297	1309.26
VAE-BM-DEC	0.737	0.450	0.452	0.450	0.443	0.458	0.450	0.737	0.033	4.683	319.99

Table 39: Full clustering metrics, GNews-H. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.243	0.495	–	–	–	–	–	0.307	–	–	–
HiCOT	0.041	0.009	–	–	–	–	–	0.041	–	–	–
BERTopic	0.784 ^{L,U}	0.916 ^{L,U}	–	–	–	–	–	0.917 ^{L,U}	–	–	–
SBERT-MiniLM	0.661	0.866	–	–	–	–	–	0.848	–	–	–
SBERT-GTE	0.706	0.884	–	–	–	–	–	0.870	–	–	–
VAE-BM	0.674	0.866	0.638	0.840	0.892	0.841	0.866	0.855	0.149	2.172	93.43
VAE-BM-PoE	0.708	0.861	0.649	0.834	0.882	0.840	0.861	0.834	0.246	1.644	209.58
VAE-BM-DEC	0.509	0.717	0.487	0.669	0.720	0.714	0.717	0.604	0.061	2.782	96.39

Table 40: Full clustering metrics, IMDB-H. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.595	0.027	–	–	–	–	–	0.595	–	–	–
HiCOT	0.505	0.001	–	–	–	–	–	0.505	–	–	–
BERTopic	0.632 ^{L,U}	0.057 ^{L,U}	–	–	–	–	–	0.632 ^{L,U}	–	–	–
SBERT-MiniLM	0.555	0.010	–	–	–	–	–	0.555	–	–	–
SBERT-GTE	0.832	0.355	–	–	–	–	–	0.832	–	–	–
VAE-BM	0.790	0.258	0.335	0.258	0.258	0.258	0.258	0.790	0.079	3.398	4183.82
VAE-BM-PoE	0.757	0.201	0.265	0.201	0.201	0.201	0.201	0.757	0.050	4.302	2625.90
VAE-BM-DEC	0.793	0.265	0.344	0.265	0.265	0.265	0.265	0.793	0.064	3.783	3409.53

Table 41: Full clustering metrics, SearchSnip.-H. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.300	0.174	–	–	–	–	–	0.351	–	–	–
HiCOT	0.454	0.378	–	–	–	–	–	0.534	–	–	–
BERTopic	0.765 ^{L,U}	0.628 ^{L,U}	–	–	–	–	–	0.797 ^{L,U}	–	–	–
SBERT-MiniLM	0.742	0.595	–	–	–	–	–	0.799	–	–	–
SBERT-GTE	0.779	0.644	–	–	–	–	–	0.838	–	–	–
VAE-BM	0.658	0.419	0.401	0.419	0.426	0.412	0.419	0.677	0.034	4.076	227.73
VAE-BM-PoE	0.812	0.614	0.612	0.614	0.621	0.608	0.614	0.812	0.078	2.716	483.58
VAE-BM-DEC	0.537	0.323	0.287	0.322	0.327	0.318	0.323	0.590	0.025	4.831	191.44

Table 42: Full clustering metrics, IMDB. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.591	0.024	–	–	–	–	–	0.591	–	–	–
HiCOT	0.504	0.001	–	–	–	–	–	0.504	–	–	–
BERTopic	0.583 ^{L,U}	0.020 ^{L,U}	–	–	–	–	–	0.583 ^{L,U}	–	–	–
SBERT-MiniLM	0.565	0.013	–	–	–	–	–	0.565	–	–	–
SBERT-GTE	0.929	0.635	–	–	–	–	–	0.929	–	–	–
VAE-BM	0.632	0.051	–	–	–	–	–	0.632	–	–	–

Table 43: Full clustering metrics, M10. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.300	0.181	–	–	–	–	–	0.318	–	–	–
HiCOT	0.259	0.114	–	–	–	–	–	0.279	–	–	–
BERTopic	0.643 ^{L,U}	0.494 ^{L,U}	–	–	–	–	–	0.654 ^{L,U}	–	–	–
SBERT-MiniLM	0.680	0.529	–	–	–	–	–	0.714	–	–	–
SBERT-GTE	0.701	0.555	–	–	–	–	–	0.730	–	–	–

Table 44: Full clustering metrics, Pascal-FI.. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.186	0.198	–	–	–	–	–	0.196	–	–	–
HiCOT	0.207	0.178	–	–	–	–	–	0.214	–	–	–
BERTopic	0.398 ^{L,U}	0.423 ^{L,U}	–	–	–	–	–	0.409 ^{L,U}	–	–	–
SBERT-MiniLM	0.364	0.387	–	–	–	–	–	0.386	–	–	–
SBERT-GTE	0.354	0.407	–	–	–	–	–	0.403	–	–	–

Table 45: Full clustering metrics, SearchSnip.. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.333	0.178	–	–	–	–	–	0.366	–	–	–
HiCOT	0.674	0.464	–	–	–	–	–	0.681	–	–	–
BERTopic	0.767 ^{L,U}	0.634 ^{L,U}	–	–	–	–	–	0.820 ^{L,U}	–	–	–
SBERT-MiniLM	0.745	0.598	–	–	–	–	–	0.802	–	–	–
SBERT-GTE	0.776	0.645	–	–	–	–	–	0.836	–	–	–
VAE-BM	0.654	0.458	–	–	–	–	–	0.725	–	–	–

Table 46: Full clustering metrics, StackOv.. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.255	0.240	–	–	–	–	–	0.269	–	–	–
HiCOT	0.430	0.392	–	–	–	–	–	0.440	–	–	–
BERTopic	0.762 ^{L,U}	0.711 ^{L,U}	–	–	–	–	–	0.762 ^{L,U}	–	–	–
SBERT-MiniLM	0.722	0.701	–	–	–	–	–	0.726	–	–	–
SBERT-GTE	0.846	0.824	–	–	–	–	–	0.846	–	–	–
VAE-BM	0.676	0.608	–	–	–	–	–	0.697	–	–	–

Table 47: Full clustering metrics, Tweet. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.477	0.637	–	–	–	–	–	0.559	–	–	–
HiCOT	0.161	0.152	–	–	–	–	–	0.168	–	–	–
BERTopic	0.604 ^{L,U}	0.876 ^{L,U}	–	–	–	–	–	0.924^{L,U}	–	–	–
SBERT-MiniLM	0.648	0.875	–	–	–	–	–	0.912	–	–	–
SBERT-GTE	<u>0.640</u>	0.881	–	–	–	–	–	<u>0.922</u>	–	–	–
VAE-BM	0.618	0.847	–	–	–	–	–	0.876	–	–	–

Table 48: Full clustering metrics, TwEval-Emo. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	<u>0.393</u>	0.047	–	–	–	–	–	0.465	–	–	–
HiCOT	<u>0.284</u>	0.005	–	–	–	–	–	0.419	–	–	–
BERTopic	0.377 ^{L,U}	0.073 ^{L,U}	–	–	–	–	–	<u>0.471^{L,U}</u>	–	–	–
SBERT-MiniLM	0.380	0.075	–	–	–	–	–	0.471	–	–	–
SBERT-GTE	0.454	0.173	–	–	–	–	–	0.605	–	–	–

Table 49: Full clustering metrics, TwEval-Sent. Lower is better for DB only; higher is better for every other metric. **Bold/underline** = best/second-best.

Model	ACC	NMI	ARI	AMI	Hom.	Comp.	V-m.	Pur.	Sil.	DB	CH
FASTopic	0.428	0.043	–	–	–	–	–	0.486	–	–	–
HiCOT	0.419	0.023	–	–	–	–	–	0.477	–	–	–
BERTopic	<u>0.456^{L,U}</u>	<u>0.058^{L,U}</u>	–	–	–	–	–	0.481 ^{L,U}	–	–	–
SBERT-MiniLM	0.464	0.068	–	–	–	–	–	0.500	–	–	–
SBERT-GTE	0.429	0.036	–	–	–	–	–	<u>0.490</u>	–	–	–